Chapter 2

Detecting Clusters Graphically

2.1 Introduction

Graphical views of multivariate data are important in all aspects of their analysis. In general terms, graphical displays of multivariate data can provide insights into the structure of the data, and in particular, from the point of view of this book, they can be useful for suggesting that the data may contain clusters and consequently that some formal method of cluster analysis might usefully be applied to the data. The usefulness of graphical displays in this context arises from the power of the human visual system in detecting patterns, and a fascinating account of how human observers draw perceptually coherent clusters out of fields of dots is given in Feldman (1995). However, the following caveat from the late Carl Sagan should be kept in mind.

> Humans are good at discerning subtle patterns that are really there, but equally so at imagining them when they are altogether absent.

In this chapter we describe a number of relatively simple, *static* graphical techniques that are often useful for providing evidence for or against possible cluster structure in the data. Most of the methods are based on an examination of either *direct* univariate or bivariate marginal plots of the multivariate data (i.e. plots obtained using the original variables), or *indirect* one- or two-dimensional 'views' of the data obtained from the application to the data of a suitable dimension-reduction technique, for example principal components analysis. For an account of *dynamic* graphical methods that may help in uncovering clusters in high-dimensional data, see Cook and Swayne (2007).

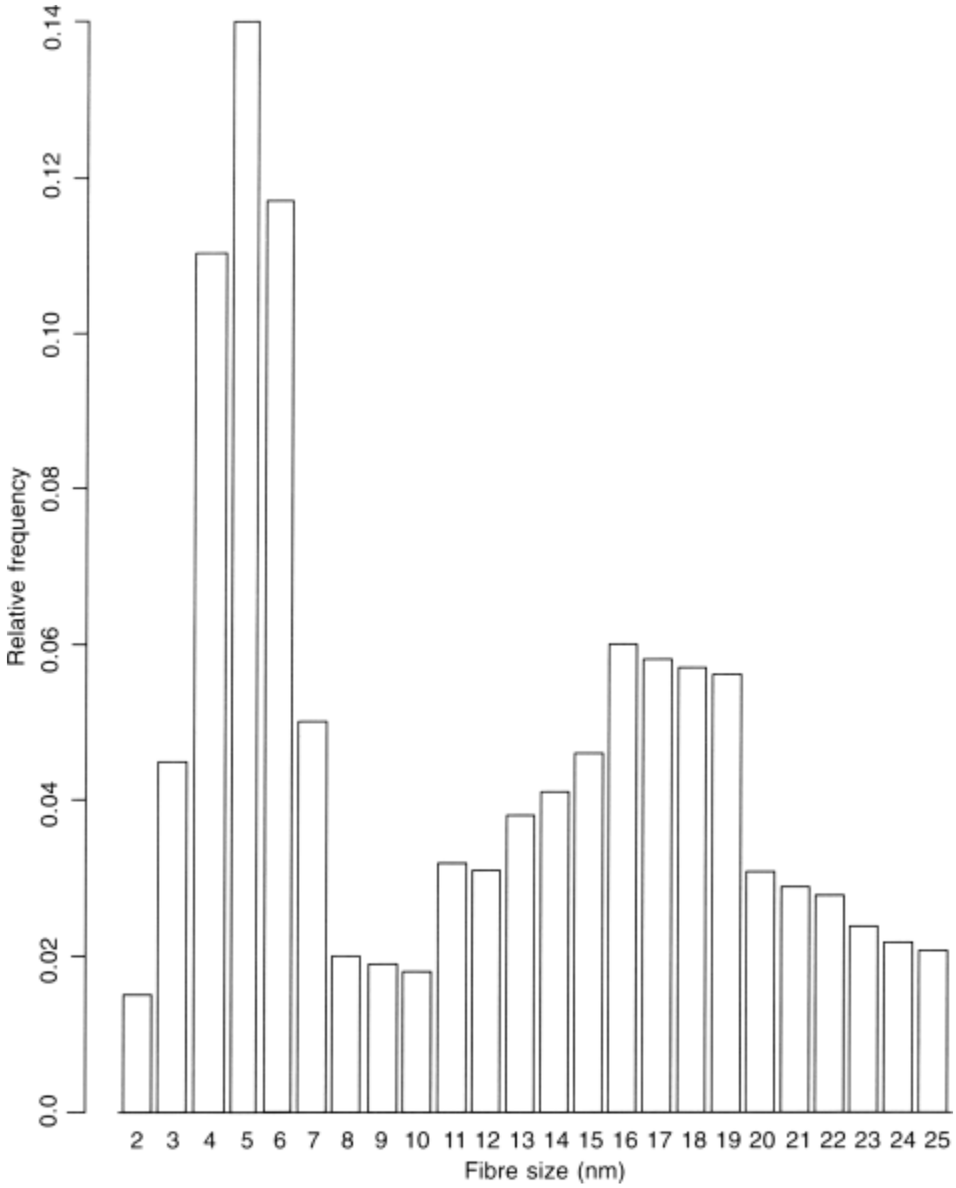
2.2 Detecting Clusters with Univariate and Bivariate Plots of Data

It is generally argued that a unimodal distribution corresponds to a homogeneous, unclustered population and, in contrast, that the existence of several distinct modes indicates a heterogeneous, clustered population, with each mode corresponding to a cluster of observations. Although this is well known not to be universally true (see, for example, Behboodian, 1970), the general thrust of the methods to be discussed in this section is that the presence of some degree of multimodality in the data is relatively strong evidence in favour of some type of cluster structure. There are a number of formal tests for the presence of distinct modes in data that may sometimes be useful; for example, those suggested by Hartigan and Hartigan (1985), Good and Gaskins (1980), Silverman (1981, 1983) and Cheng and Hall (1998). But here we shall not give details of these methods, preferring to concentrate on a rather more informal 'eye-balling' approach to the problem of mode detection using suitable one- and two-dimensional plots of the data.

2.2.1 Histograms

The humble histogram is often a useful first step in the search for modes in data, particularly, of course, if the data are univariate. Figure 2.1, for example, shows a histogram of myelinated lumbosacral ventral root fibre sizes from a kitten of a particular age. The distribution is distinctly bimodal, suggesting the presence of two relatively distinct groups of observations in the data. Here it is known that the first mode is associated with axons of gamma neurones and the second with alpha neurones.

Figure 2.1 Histogram of myelinated lumbosacral ventral root fibre sizes from a kitten of a particular age.



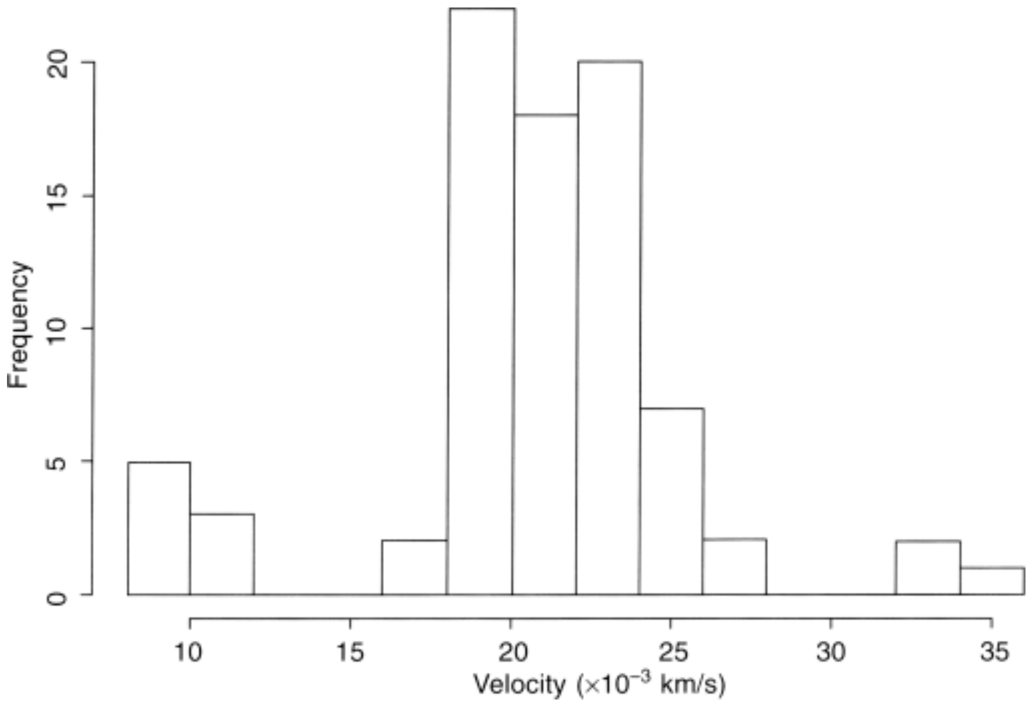
Now consider the data shown in Table 2.1, which gives the velocities of 82 galaxies from six well-separated conic sections of space. The data are intended to shed light on whether or not the observable universe contains superclusters of galaxies surrounded by large voids. The evidence for the existence of the superclusters would be in the multimodality of the distribution of velocities. A histogram of the data does give some evidence of such multimodality – see Figure 2.2.

Table 2.1 Velocities of 82 galaxies (km/s)

9172	9558	10406	18419	18927	19330	19440	19541	19846	19914	19989
20179	20221	20795	20875	21492	21921	22209	22314	22746	22914	23263
23542	23711	24289	24990	26995	34279	9350	9775	16084	18552	19052
19343	19473	19547	19856	19918	20166	20196	20415	20821	20986	21701
21960	22242	22374	22747	23206	23484	23666	24129	24366	25633	32065
9483	10227	16170	18600	19070	19349	19529	19663	19863	19973	20175
10215	20629	20846	21137	21814	22185	22249	22495	22888	23241	23538
23706	24285	24717	26960	32789						

With multivariate data, histograms can be constructed for each separate variable, as we shall in a later example in this chapter (see Section 2.2.4), but in general this will not be of great help in uncovering cluster structure in the data because the marginal distribution of each variable may not reflect accurately the multivariate distribution of the complete set of variables. It is, for example, possible for clustered multivariate data to give rise to unimodal histograms for some or all of the separate variables.

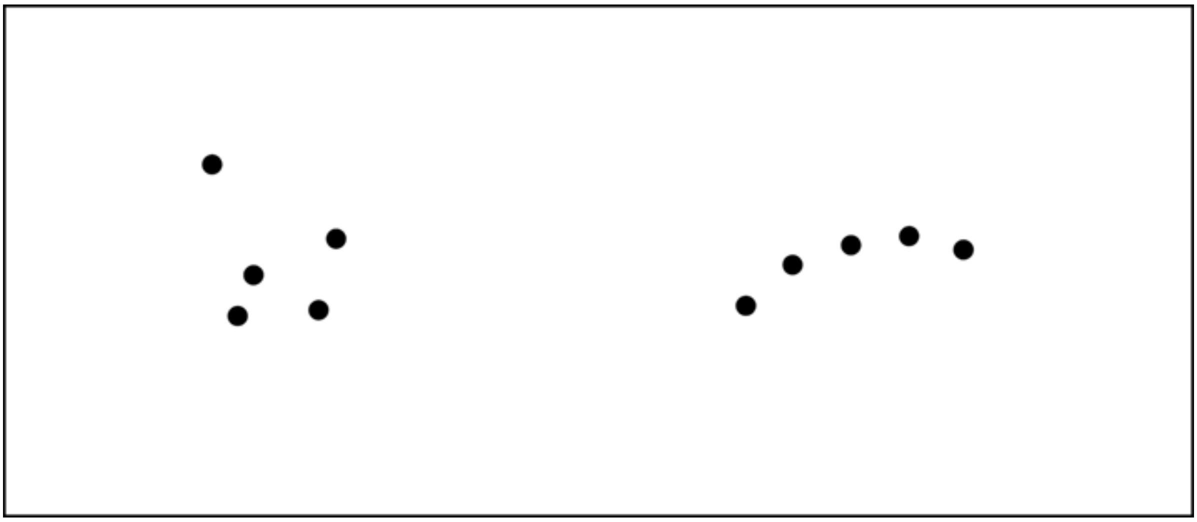
Figure 2.2 Histogram of velocities of 82 galaxies.



2.2.2 Scatterplots

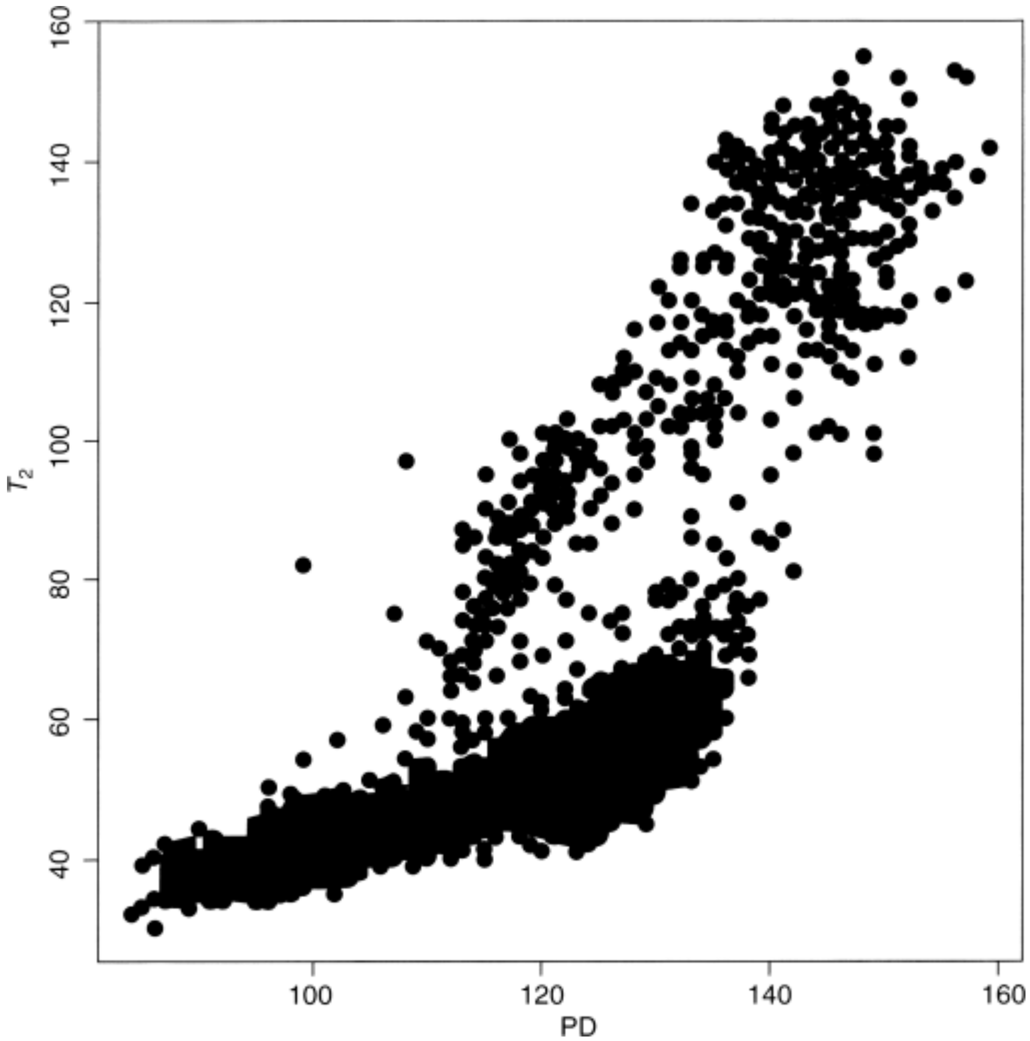
The basis of two-dimensional views of the data is the simple *xy*-scatterplot, which has been in use since at least the 18th century (see Tufte, 1983). To begin, consider the scatterplot shown in Figure 2.3. To a human observer (such as the reader), it is obvious that there are two natural groups or clusters of dots. (It is not entirely clear how a 'cluster' is recognized when displayed in the plane, although one feature of the recognition process probably involves the assessment of the relative distances between points.) This conclusion is reached with no conscious effort of thought. The relative homogeneity of each cluster of points and the degree of their separation makes the task of identifying them simple.

Figure 2.3 Scatterplot showing two distinct groups of points.



But now consider the example provided by Figure 2.4, which is a scatterplot of two measures of intensity, PD and T_2, for 2836 voxels from a functional magnetic resonance imaging (fMRI) study reported in Bullmore *et al*. (1995). *Some* cluster structure is apparent from this plot, but in parts of the diagram the 'overplotting' may hide the presence of other clusters. And such will be the case for the scatterplots of many data sets found in practice. The basic scatterplot might, it appears, need a little help if it is to reveal *all* the structure in the data.

Figure 2.4 Scatterplot of fMRI data.



2.2.3 Density Estimation

Scatterplots (and histograms) can be made to be more useful by adding a numerical estimate of the bivariate (univariate) density of the data. If we are willing to assume a particular form for the distribution of the data, for example bivariate (univariate) Gaussian, density estimation would be reduced to simply estimating the values of the density's parameters. (A particular type of parameter density function useful for modelling clustered data will be described in Chapter 6.) More commonly, however, we wish to allow the data to speak for themselves and so use one of a variety of *nonparametric estimation* procedures now available. Density estimation is now a large topic and is covered in several books including Silverman (1986), Scott (1992), Wand and Jones (1995), Simonoff (1996) and Bowman and Azzalini (1997). Here we content ourselves with a brief account of *kernel density estimates*, beginning for simplicity with estimation for the univariate situation and then moving on to the usually more interesting (at least from a clustering perspective) case of bivariate data.

Univariate Density Estimation

From the definition of a probability density, if the random variable *X* has density *f*,

(2.1) $$f(x) = \lim_{h \to 0} \frac{1}{2h} P(x-h < X < x+h).$$

For any given *h*, a naïve estimator of $P(x-h < X < x+h)$ is the proportion of the observations $X_1, X_2, \ldots, X_n$ falling in the interval $(x-h,\ x+h)$; that is

(2.2) $$\hat{f}(x) = \frac{1}{2hn}[\text{no. of } X_1, X_2, \ldots, X_n \text{ falling in } (x-h,\ x+h)].$$

If we introduce a weight function *W* given by

(2.3) $$W(x) = \left\{ \begin{array}{ll} \frac{1}{2} & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{array} \right\},$$

then the naïve estimator can be rewritten as

(2.4) $$\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{h} W\left(\frac{x-X_i}{h}\right).$$

Unfortunately, this estimator is not a continuous function and is not satisfactory for practical density estimation. It does, however, lead naturally to the kernel estimator defined by

(2.5) $$\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x-X_i}{h}\right),$$

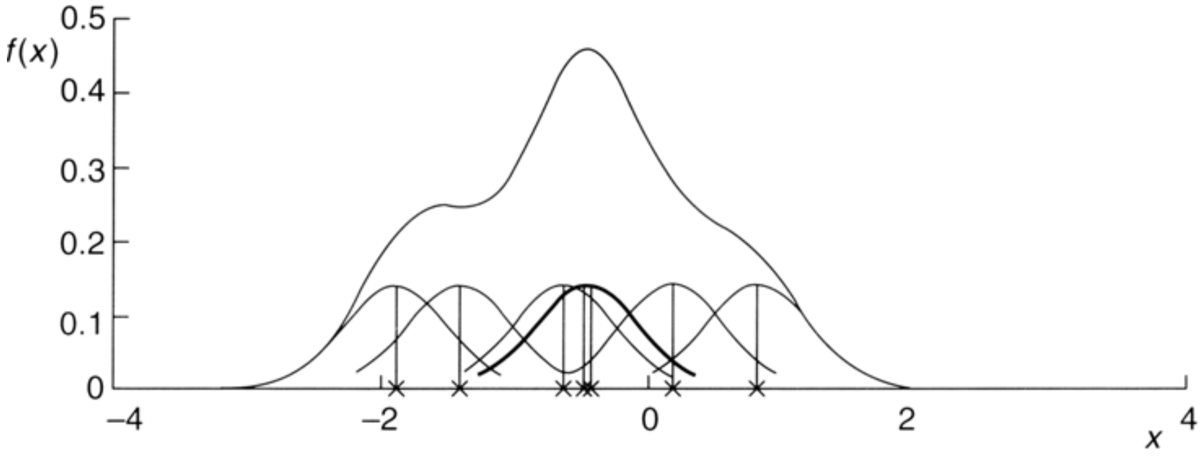
where *K* is known as the *kernel function* and *h* as the *bandwidth* or *smoothing parameter*. The kernel function must satisfy the condition

(2.6) $$\int_{-\infty}^{\infty} K(x)dx = 1.$$

Usually, but not always, the kernel function will be a symmetric density function, for example, the normal.

The kernel estimator is a sum of 'bumps' placed at the observations. The kernel function determines the shape of the bumps, while the window width *h* determines their width. Figure 2.5 (taken from Silverman, 1986) shows the individual bumps $n^{-1}h^{-1}K[(x-X_i)/h]$, as well as the estimate $\hat{f}$ obtained by adding them up.

Figure 2.5 Kernel estimate showing individual kernels.



Three commonly used kernel functions are *rectangular*, *triangular* and *Gaussian*:

- rectangular

$$K(x) = \frac{1}{2} \text{ for } |x| < 1, \ 0 \text{ otherwise} \tag{2.7}$$

- triangular

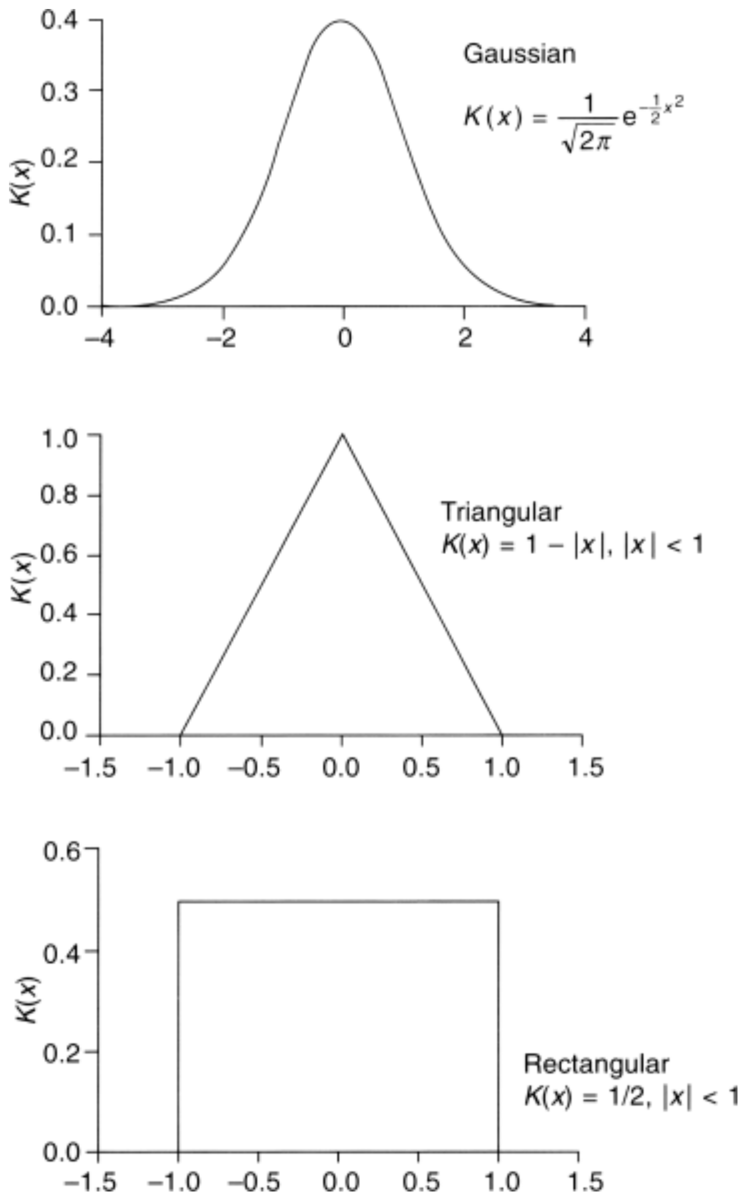
$$K(x) = 1 - |x| \text{ for } |x| < 1, \ 0 \text{ otherwise} \tag{2.8}$$

- Gaussian

$$K(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}. \tag{2.9}$$

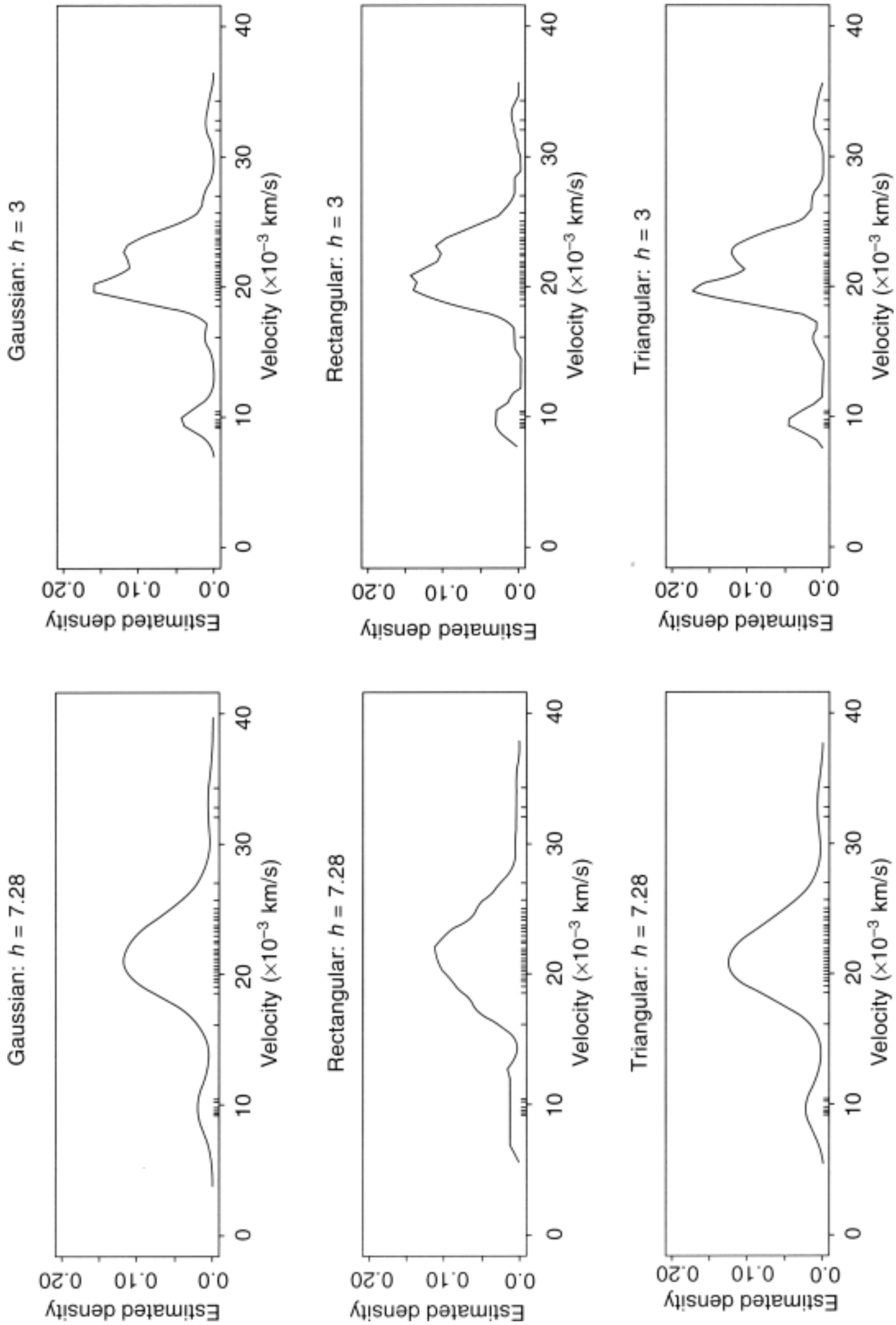
The three kernel functions are illustrated in Figure 2.6.

Figure 2.6 Three kernel functions.



As an illustration of applying univariate kernel density estimators, Figure 2.7 shows a number of estimates for the velocities of galaxies data in Table 2.1, obtained by using different kernel functions and/or different values for the bandwidth. The suggestion of multimodality and hence 'clustering' in the galaxy velocities is very strong.

Figure 2.7 Kernel density estimates for galaxy velocity data.



Bivariate Density Estimation

The kernel density estimator considered as a sum of 'bumps' centred at the observations has a simple extension to two dimensions (and similarly for more than two dimensions). The bivariate estimator for data (X_1, Y_1), $(X_2, Y_2), \ldots, (X_n, Y_n)$ is defined as

$$\hat{f}(x,y) = \frac{1}{nh_x h_y}\sum_{i=1}^{n} K\left(\frac{x-X_i}{h_x}, \frac{y-Y_i}{h_y}\right). \tag{2.10}$$

In this estimator each coordinate direction has its own smoothing parameter, h_x and h_y. An alternative is to scale the data equally for both dimensions and use a single smoothing parameter.

For bivariate density estimation a commonly used kernel function is the standard bivariate normal density

$$K(x,y) = \frac{1}{2\pi}\exp\left[-\frac{1}{2}\left(x^2+y^2\right)\right]. \tag{2.11}$$

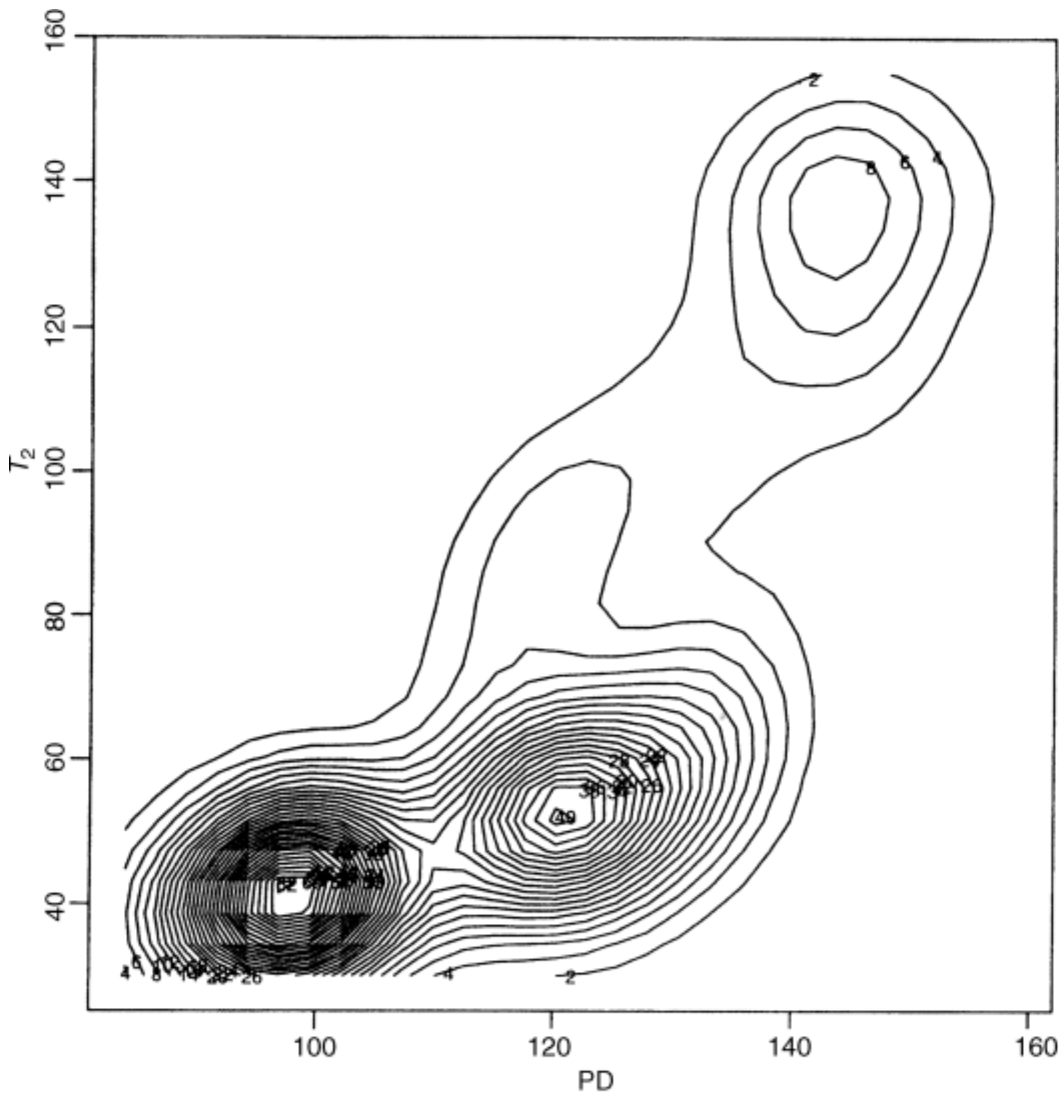
Another possibility is the bivariate Epanechikov kernel given by

$$K(x,y)\begin{cases} = \frac{2}{\pi}\left(1-x^2-y^2\right) & \text{if } x^2+y^2<1 \\ = 0 \text{ otherwise} & \end{cases}. \tag{2.12}$$

According to Venables and Ripley (1999), the bandwidth should be chosen to be proportional to $n^{-\frac{1}{5}}$; unfortunately the constant of proportionality depends on the unknown density. The tricky problem of bandwidth estimation is considered in detail in Silverman (1986).

Returning to the fMRI data introduced in the previous subsection and plotted in Figure 2.4, we will now add the contours of the estimated bivariate density of the data to this scatterplot, giving the diagram shown in Figure 2.8. The enhanced scatterplot gives evidence of at least three clusters of voxels, a structure that could not be seen in Figure 2.4. Here, three clusters makes good sense because they correspond largely to grey matter voxels, white matter voxels and cerebrospinal fluid (CSF) voxels. (A more formal analysis of these data is presented in Chapter 6.)

Figure 2.8 Scatterplot of fMRI data enhanced with estimated bivariate density.



2.2.4 Scatterplot Matrices

When we have multivariate data with three or more variables, the scatterplots of each pair of variables can still be used as the basis of an initial examination of the data for informal evidence that the data have some cluster structure, particularly if the scatterplots are arranged as a *scatterplot matrix*. A scatterplot matrix is defined as a square, symmetric grid of bivariate scatterplots (Cleveland, 1994). This grid has p rows and p columns, each one corresponding to a different one of the p variables. Each of the grid's cells show a scatterplot of two variables. Because the scatterplot matrix is symmetric about its diagonal, variable j is plotted against variable i in the ijth cell, and the same variables appear in cell ji with the x and y axes of the scatterplots interchanged. The reason for including both the upper and lower triangles of the grid, despite its seeming redundancy, is that it enables a row and a column to be visually scanned to see one variable against all others, with the scales for the one lined up along the horizontal or the vertical.

To illustrate the use of the scatterplot matrix combined with univariate and bivariate density estimation, we shall use the data on body measurements shown in Table 2.2. (There are, of course, too few *observations* here for an entirely satisfactory or convincing demonstration, but they will serve as a useful example.) Figure 2.9 is a scatterplot matrix of the data in which each component scatterplot is enhanced with the estimated bivariate density of the two variables, and the diagonal panels display the histograms for each separate variable. There is clear evidence of the presence of two separate groups, particularly in the waist–hips measurements scatterplot. In this case the explanation is simple – the data contain measurements on both males and females.

Table 2.2 Body measurements data (inches).

Subject	Chest	Waist	Hips
1	34	30	32
2	37	32	37
3	38	30	36
4	36	33	39
5	38	29	33
6	43	32	38
7	40	33	42
8	38	30	40
9	40	30	37
10	41	32	39
11	36	24	35
12	36	25	37
13	34	24	37
14	33	22	34
15	36	26	38
16	37	26	37
17	34	25	38
18	36	26	37
19	38	28	40
20	35	23	35

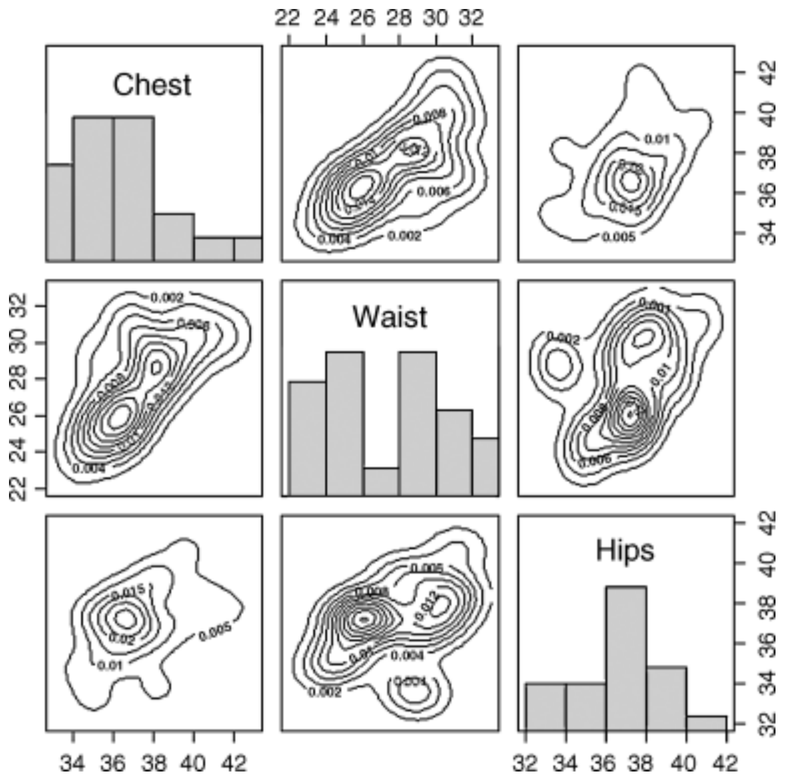
As a second example of using the scatterplot matrix combined with bivariate density estimation, we shall use the data shown in Table 2.3 collected in a study of air pollution in 41 US cities. The variables recorded are as follows:

SO_2	Sulphur dioxide content of air in micrograms per cubic metre;
TEMP	Average annual temperature (°F);
MANUF	Number of manufacturing enterprises employing 20 or more workers;
POP	Population size (1970 census) in thousands;
WIND	Average wind speed in miles per hour;
PRECIP	Average annual precipitation in inches;
DAYS	Average number of days with precipitation per year.

Table 2.3 Air pollution in US cities.

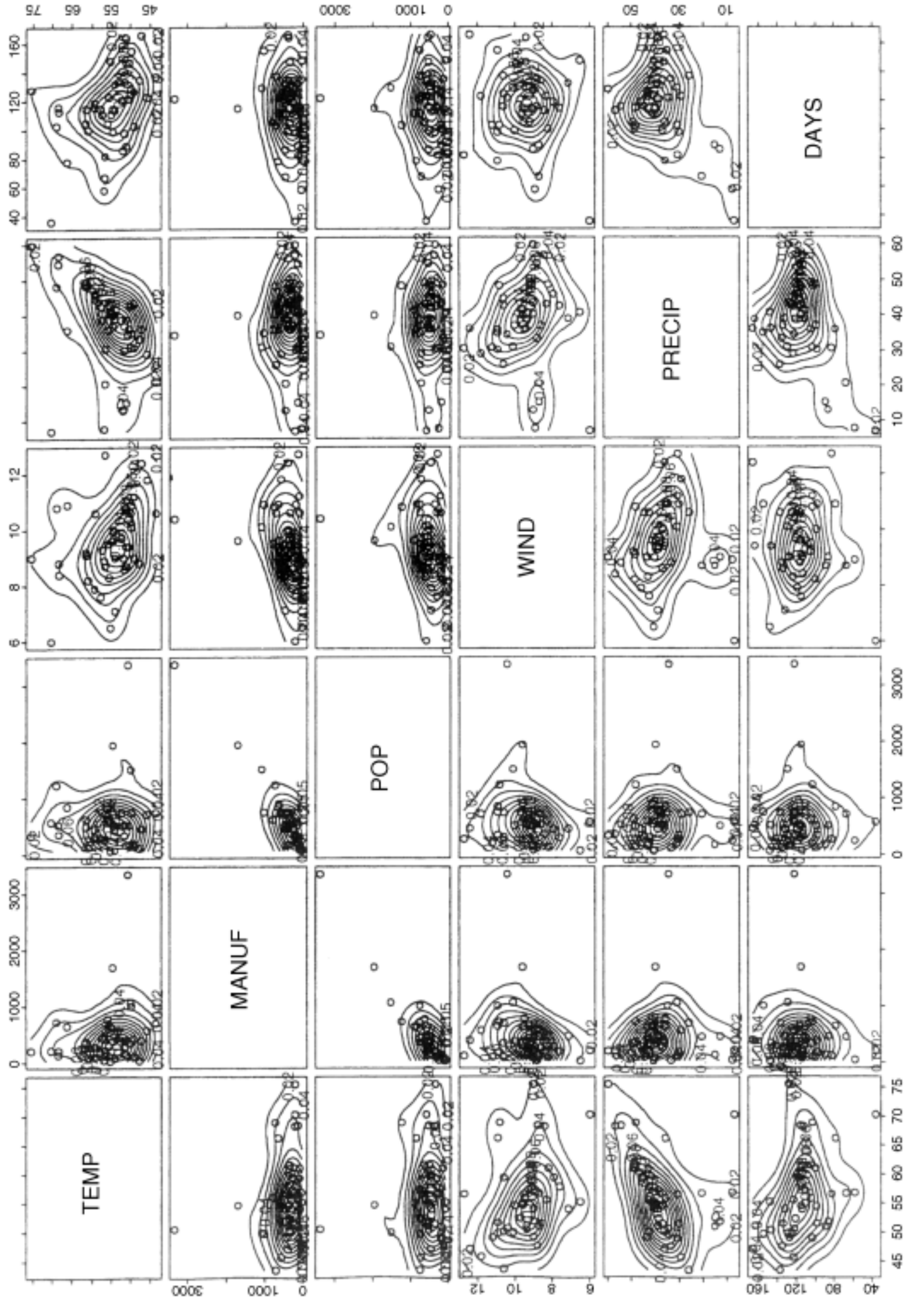
City	SO2	TEMP	MANUF	POP	WIND	PRECIP	DAYS
Phoenix	10	70.3	213	582	6.0	7.05	36
Little Rock	13	61.0	91	132	8.2	48.52	100
San Francisco	12	56.7	453	716	8.7	20.66	67
Denver	17	51.9	454	515	9.0	12.95	86
Hartford	56	49.1	412	158	9.0	43.37	127
Wilmington	36	54.0	80	80	9.0	40.25	114
Washington	29	57.3	434	757	9.3	38.89	111
Jacksonville	14	68.4	136	529	8.8	54.47	116
Miami	10	75.5	207	335	9.0	59.80	128
Atlanta	24	61.5	368	497	9.1	48.34	115
Chicago	110	50.6	3344	3369	10.4	34.44	122
Indianapolis	28	52.3	361	746	9.7	38.74	121
Des Moines	17	49.0	104	201	11.2	30.85	103
Wichita	8	56.6	125	277	12.7	30.58	82
Louisville	30	55.6	291	593	8.3	43.11	123
New Orleans	9	68.3	204	361	8.4	56.77	113
Baltimore	47	55.0	625	905	9.6	41.31	111
Detroit	35	49.9	1064	1513	10.1	30.96	129
Minneapolis	29	43.5	699	744	10.6	25.94	137
Kansas	14	54.5	381	507	10.0	37.00	99
St Louis	56	55.9	775	622	9.5	35.89	105
Omaha	14	51.5	181	347	10.9	30.18	98
Albuquerque	11	56.8	46	244	8.9	7.77	58
Albany	46	47.6	44	116	8.8	33.36	135
Buffalo	11	47.1	391	463	12.4	36.11	166
Cincinnati	23	54.0	462	453	7.1	39.04	132
Cleveland	65	49.7	1007	751	10.9	34.99	155
Columbus	26	51.5	266	540	8.6	37.01	134
Philadelphia	69	54.6	1692	1950	9.6	39.93	115
Pittsburgh	61	50.4	347	520	9.4	36.22	147
Providence	94	50.0	343	179	10.6	42.75	125
Memphis	10	61.6	337	624	9.2	49.10	105
Nashville	18	59.4	275	448	7.9	46.00	119
Dallas	9	66.2	641	844	10.9	35.94	78
Houston	10	68.9	721	1233	10.8	48.19	103
Salt Lake City	28	51.0	137	176	8.7	15.17	89
Norfolk	31	59.3	96	308	10.6	44.68	116
Richmond	26	57.8	197	299	7.6	42.59	115
Seattle	29	51.1	379	531	9.4	38.79	164
Charleston	31	55.2	35	71	6.5	40.75	148
Milwaukee	16	45.7	569	717	11.8	29.07	123

Figure 2.9 Scatterplot matrix of body measurements enhanced by bivariate density estimates and histograms.



Here we shall use the six climate and human ecology variables to assess whether there is any evidence that there are groups of cities with similar profiles. (The sulphur dioxide content of the air will be used in Chapter 5 to aid in evaluating a more detailed cluster analysis of these data.) A scatterplot matrix of the data in which each component scatterplot is enhanced with the estimated bivariate density of the two variables is shown in Figure 2.10. The plot gives little convincing evidence of any group structure in the data. The most obvious feature of the data is the presence of several 'outlier' observations. As such observations can be a problem for some methods of cluster analysis, identification of them prior to applying these methods is often very useful.

Figure 2.10 Scatterplot matrix of air pollution data in which each panel is enhanced with an estimated bivariate density.



2.3 Using Lower-Dimensional Projections of Multivariate Data for Graphical Representations

Scatterplots and, to some extent, scatterplot matrices are more useful for exploring multivariate data for the presence of clusters when there are only a relatively small number of variables. When the number of variables, p, is moderate to large (as it will be for many multivariate data sets encountered in practice), the two-dimensional marginal views of the data suffer more and more from the fact that they do not necessarily reflect the true nature of the structure present in p dimensions. But all is not necessarily lost, and scatterplots etc. may still be useful after the data have been projected into a small number of dimensions in some way that preserves their multivariate structure as fully as possible. There are a number of possibilities, of which the most common (although not necessarily the most useful) is *principal components analysis*.

2.3.1 Principal Components Analysis of Multivariate Data

Principal components analysis is a method for transforming the variables in a multivariate data set into new variables which are uncorrelated with each other and which account for decreasing proportions of the total variance of the original variables. Each new variable is defined as a particular linear combination of the original variables. Full accounts of the method are given in Everitt and Dunn (2001) and Jackson (2003), but in brief:

- The first principal component, y_1, is defined as the linear combination of the original variables, $x_1, x_2, \ldots, x_p$, that accounts for the maximal amount of the variance of the x variables amongst all such linear combinations.
- The second principal component, y_2, is defined as the linear combination of the original variables that accounts for a maximal amount of the remaining variance subject to being uncorrelated with y_1. Subsequent components are defined similarly.
- So principal components analysis finds new variables $y_1, y_2, \ldots, y_p$ defined as follows:

$$\begin{aligned} y_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1p}x_p \\ y_2 &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2p}x_p \\ &\vdots \\ y_p &= a_{p1}x_1 + a_{p2}x_2 + \cdots + a_{pp}x_p, \end{aligned} \tag{2.13}$$

  with the coefficients being chosen so that $y_1, y_2, \ldots, y_p$ account for decreasing proportions of the variance of the x variables and are uncorrelated. (Because the variance can be scaled by rescaling the combination defining a principal component, the coefficients defining each principal component are such that their sums of squares equal one.)
- The coefficients are found as the eigenvectors of the sample covariance matrix **S** or, more commonly, the sample correlation matrix **R** when the variables are on very different scales. It should be noted that there is, in general, no simple relationship between the components found from these two matrices. (A similar scaling decision is also often a problem in cluster analysis, as we shall see in Chapter 3).
- The variances of $y_1, y_2, \ldots, y_p$ are given by the eigenvalues of **S** or **R**. Where the first few components account for a large proportion of the variance of the observed variables, it is generally assumed that they provide a parsimonious summary of the original data, useful perhaps in later analysis.

In a clustering context, principal components analysis provides a means of projecting the data into a lower dimensional space, so as to make visual inspection hopefully more informative. The points of the q-dimensional projection ($q < p$) onto the first q principal components lie in a q-dimensional space, and this is the best-fitting q-dimensional space as measured by the sum of the squares of the distances from the data points to their projections into this space.

The use of principal components analysis in the context of clustering will be illustrated on the data shown in Table 2.4, which gives the chemical composition in terms of nine oxides as determined by atomic absorption spectrophotometry, of 46 examples of Romano-British pottery (Tubb *et al*., 1980). Here there are nine variables, recorded for each piece of pottery.

The scatterplot matrix of the data shown in Figure 2.11 does show some evidence of clustering, but it is quite difficult to see clearly any detail of the possible structure in the data.

Table 2.4 Results of chemical analysis of Roman-British pottery (Tubb *et al.,* 1980)

Sample number	Chemical component								
	Al_2O_3	Fe_2O_3	MgO	CaO	Na_2O	K_2O	TiO_2	MnO	BaO
1	18.8	9.52	2.00	0.79	0.40	3.20	1.01	0.077	0.015
2	16.9	7.33	1.65	0.84	0.40	3.05	0.99	0.067	0.018
3	18.2	7.64	1.82	0.77	0.40	3.07	0.98	0.087	0.014
4	16.9	7.29	1.56	0.76	0.40	3.05	1.00	0.063	0.019
5	17.8	7.24	1.83	0.92	0.43	3.12	0.93	0.061	0.019
6	18.8	7.45	2.06	0.87	0.25	3.26	0.98	0.072	0.017
7	16.5	7.05	1.81	1.73	0.33	3.20	0.95	0.066	0.019
8	18.0	7.42	2.06	1.00	0.28	3.37	0.96	0.072	0.017
9	15.8	7.15	1.62	0.71	0.38	3.25	0.93	0.062	0.017
10	14.6	6.87	1.67	0.76	0.33	3.06	0.91	0.055	0.012
11	13.7	5.83	1.50	0.66	0.13	2.25	0.75	0.034	0.012
12	14.6	6.76	1.63	1.48	0.20	3.02	0.87	0.055	0.016
13	14.8	7.07	1.62	1.44	0.24	3.03	0.86	0.080	0.016
14	17.1	7.79	1.99	0.83	0.46	3.13	0.93	0.090	0.020
15	16.8	7.86	1.86	0.84	0.46	2.93	0.94	0.094	0.020
16	15.8	7.65	1.94	0.81	0.83	3.33	0.96	0.112	0.019
17	18.6	7.85	2.33	0.87	0.38	3.17	0.98	0.081	0.018
18	16.9	7.87	1.83	1.31	0.53	3.09	0.95	0.092	0.023
19	18.9	7.58	2.05	0.83	0.13	3.29	0.98	0.072	0.015
20	18.0	7.50	1.94	0.69	0.12	3.14	0.93	0.035	0.017
21	17.8	7.28	1.92	0.81	0.18	3.15	0.90	0.067	0.017
22	14.4	7.00	4.30	0.15	0.51	4.25	0.79	0.160	0.019
23	13.8	7.08	3.43	0.12	0.17	4.14	0.77	0.144	0.020
24	14.6	7.09	3.88	0.13	0.20	4.36	0.81	0.124	0.019
25	11.5	6.37	5.64	0.16	0.14	3.89	0.69	0.087	0.009
26	13.8	7.06	5.34	0.20	0.20	4.31	0.71	0.101	0.021
27	10.9	6.26	3.47	0.17	0.22	3.40	0.66	0.109	0.010
28	10.1	4.26	4.26	0.20	0.18	3.32	0.59	0.149	0.017
29	11.6	5.78	5.91	0.18	0.16	3.70	0.65	0.082	0.015
30	11.1	5.49	4.52	0.29	0.30	4.03	0.63	0.080	0.016
31	13.4	6.92	7.23	0.28	0.20	4.54	0.69	0.163	0.017
32	12.4	6.13	5.69	0.22	0.54	4.65	0.70	0.159	0.015
33	13.1	6.64	5.51	0.31	0.24	4.89	0.72	0.094	0.017
34	11.6	5.39	3.77	0.29	0.06	4.51	0.56	0.110	0.015
35	11.8	5.44	3.94	0.30	0.04	4.64	0.59	0.085	0.013
36	18.3	1.28	0.67	0.03	0.03	1.96	0.65	0.001	0.014
37	15.8	2.39	0.63	0.01	0.04	1.94	1.29	0.001	0.014
38	18.0	1.50	0.67	0.01	0.06	2.11	0.92	0.001	0.016
39	18.0	1.88	0.68	0.01	0.04	2.00	1.11	0.006	0.022
40	20.8	1.51	0.72	0.07	0.10	2.37	1.26	0.002	0.016
41	17.7	1.12	0.56	0.06	0.06	2.06	0.79	0.001	0.013
42	18.3	1.14	0.67	0.06	0.05	2.11	0.89	0.006	0.019
43	16.7	0.92	0.53	0.01	0.05	1.76	0.91	0.004	0.013
44	14.8	2.74	0.67	0.03	0.05	2.15	1.34	0.003	0.015
45	19.1	1.64	0.60	0.10	0.03	1.75	1.04	0.007	0.018

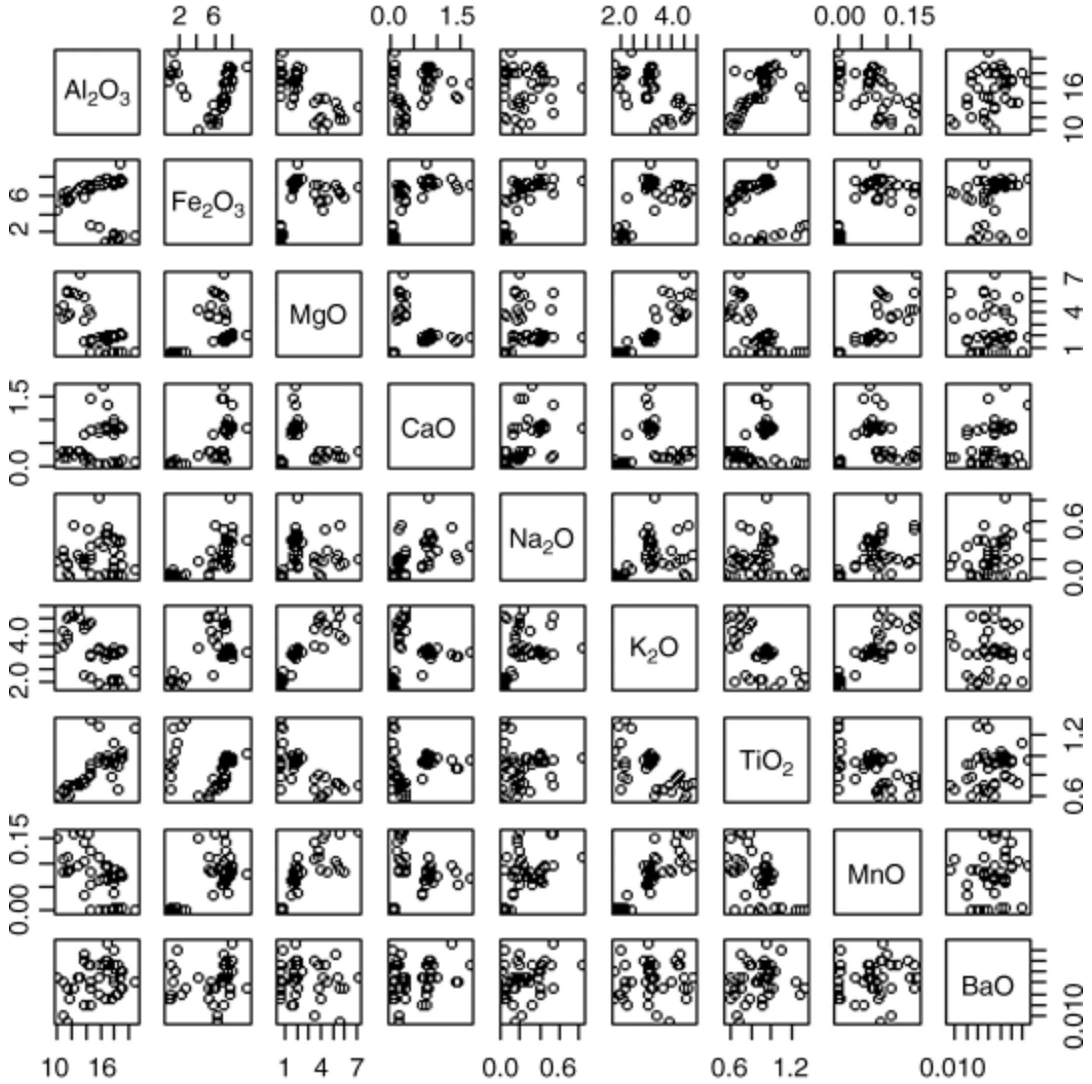
The results of a principal components analysis on the correlation matrix of the pottery data are shown in Table 2.5. The first two components have eigenvalues greater than one, and the third component has an eigenvalue very close to one, and so we will use these three components to represent the data graphically. (Here we shall not spend time trying to interpret the components.) A scatterplot matrix of the first three component scores for each piece of pottery is shown in Figure 2.12. Each separate scatterplot is enhanced with a contour plot of the appropriate estimated bivariate density. The diagram gives strong evidence that there are at least three clusters in the data. Here an explanation for the cluster structure is found by looking at the regions from which the pots arise; the three clusters correspond to pots from three different regions – see Everitt (2005).

Table 2.5 Results of a principal components analysis on the Romano-British pottery data.

Oxide	Component 1	Component 2	Component 3
Al_2O_3	0.35	0.33	−0.12
Fe_2O_3	−0.33	0.40	0.26
MgO	−0.44	−0.19	−0.15
CaO	-----	0.50	0.48
Na_2O	−0.22	0.46	-----
K_2O	−0.46	-----	−0.10
TiO_2	0.34	0.30	-----
MnO	−0.46	-----	−0.14
BaO	-----	0.38	−0.79
Component standard deviation	2.05	1.59	0.94

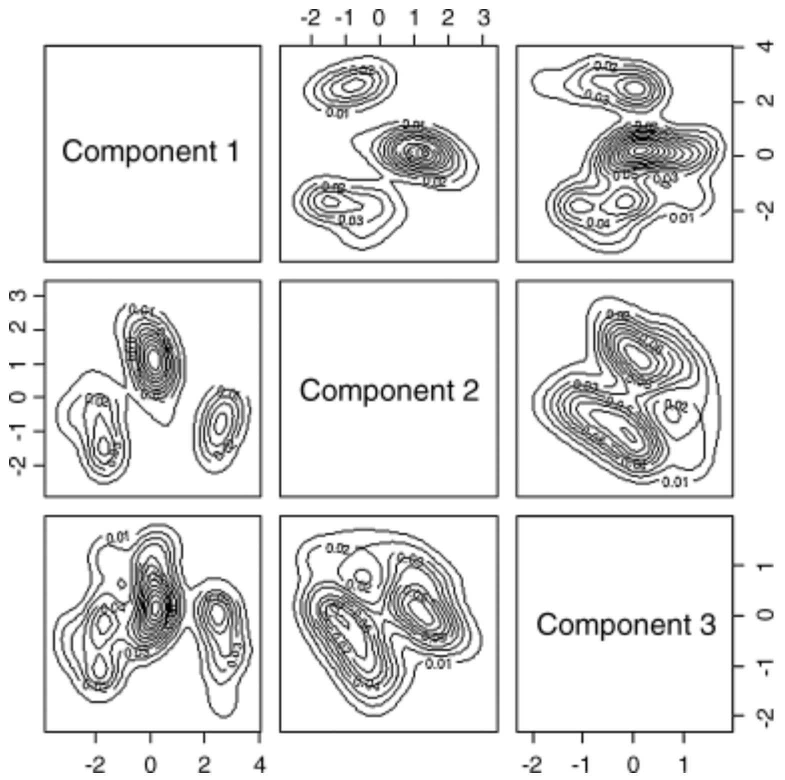
----- = the loading is very close to zero.

Figure 2.11 Scatterplot matrix of data on Romano-British pottery.



(A last word of caution is perhaps needed about using principal components in the way described above, because the principal components analysis may attribute sample correlations due to clusters as due to components. The problem is taken up in detail in Chapters 3, 6 and 7.)

Figure 2.12 Scatterplot of the first three principal components of the Romano-British pottery data enhanced by bivariate density estimates.



2.3.2 Exploratory Projection Pursuit

According to Jones and Sibson (1987),

> The term 'projection pursuit' was first used by Friedman and Tukey (1974) to name a technique for the exploratory analysis of reasonably large and reasonably multivariate data sets. Projection pursuit reveals structure in the original data by offering selected low-dimensional orthogonal projections of it for inspection.

In essence, projection pursuit methods seek a *q*-dimensional projection of the data that maximizes some measure of 'interestingness', usually for $q = 1$ or $q = 2$ so that they can be visualized. Principal components analysis is, for example, a projection pursuit method in which the index of interestingness is the proportion of total variance accounted for by the projected data. It relies for its success on the tendency for large variation to also be interestingly structured variation, a connection which is not necessary and which often fails to hold in practice. The essence of projection pursuit can be described most simply by considering only one-dimensional solutions.

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be a *p*-dimensional data set of size *n*; then the projection is specified by a *p*-dimensional vector α such that $\alpha\alpha' = 1$. (A two-dimensional projection would involve a $p \times 2$ matrix). The projected data in this case are one-dimensional data points $z_1, \ldots, z_n$, where $z_i = \alpha\mathbf{x}_i$ for $i = 1, \ldots, n$. The merit of a particular configuration $\{z_1, \ldots, z_n\}$ in expressing important structure in the data will be quantified by means of a projection index, $I_{(z_1,\ldots,z_n)}$. (Principal components reduction effectively uses sample variance as a projection index.) Most frequently the index used is a measure of the discrepancy between the density of the projected data and the density of an 'uninteresting' projection, for example the standard normal, but many other indices of 'interestingness' have been suggested. Details are given in Jones and Sibson (1987), Ripley (1996) and Sun (1998).

Once an index is chosen, a projection is found by numerically maximizing the index over the choice of projection; that is, a *q*-dimensional projection is determined by a $p \times q$ orthogonal matrix. With *q* small (usually one or two), this may appear to imply a relatively simple optimization task, but Ripley (1996) points out why this is not always the case:

- The index is often very sensitive to the projection directions, and good views may occur with sharp and well-separated peaks in the optimization space.
- The index may be very sensitive to small changes in the data configuration and so have very many local maxima.

As a result of such problems, Friedman, in the discussion of Jones and Sibson (1987), reflects thus:

> It has been my experience that finding the substantive minima of a projection index is a difficult problem, and that simple gradient-guided methods (such as steepest descent) are generally inadequate. The power of a projection pursuit procedure depends crucially on the reliability and thoroughness of the numerical optimizer.

Ripley (1996) supports this view and suggests that it may be necessary to try many different starting points, some of which may reveal projections with large values of the projection index. Posse (1990, 1995), for example, considers an almost random search which he finds to be superior to the optimization methods of Jones and Sibson (1987) and Friedman (1987).

As an illustration of the application of projection pursuit, we will describe an example given in Ripley (1996). This involves data from a study reported in Campbell and Mahon (1974) on rock crabs of the genus *Leptograpsus*. Each specimen has measurements on the following five variables:

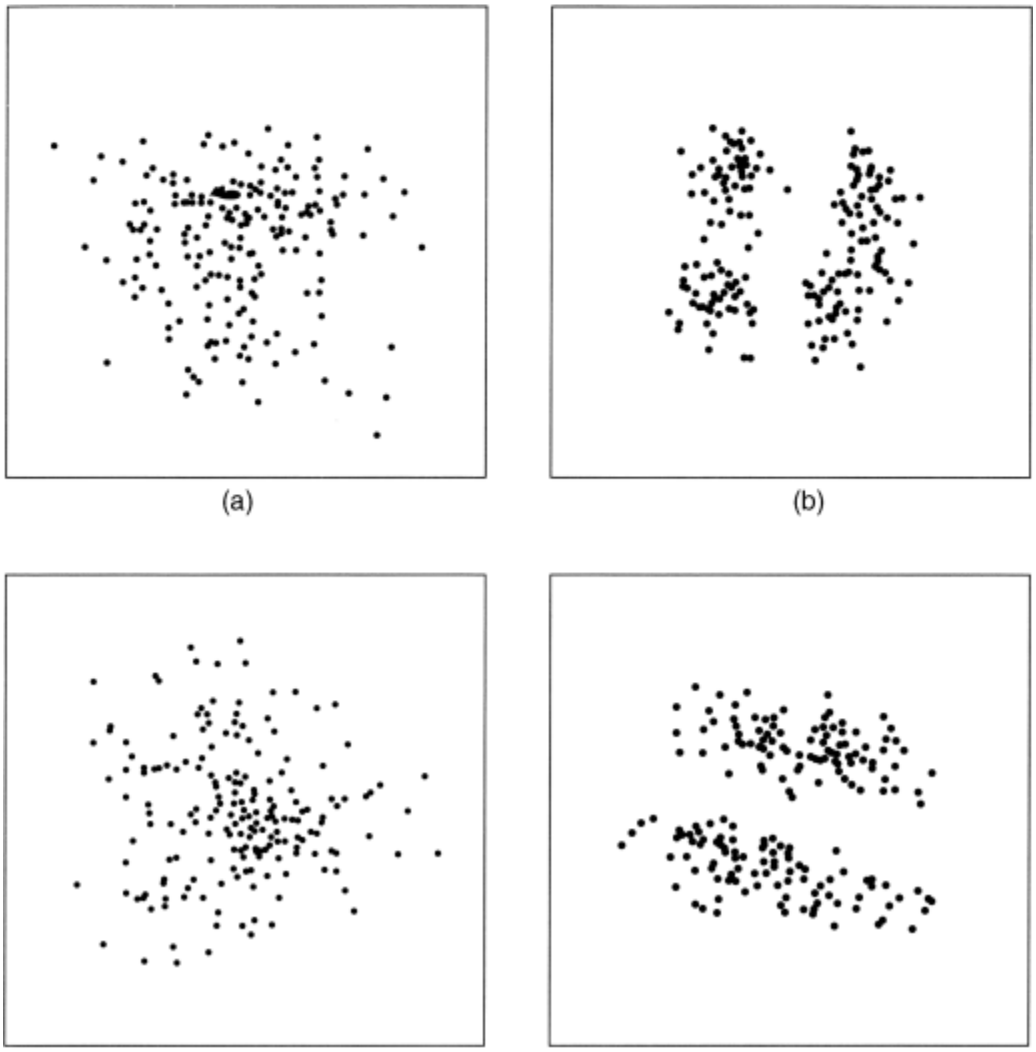
FL	width of frontal lip
RW	rear width
CL	length along midline
CW	maximum width of the carapace
BD	body depth.

(All measurements were in mm).

Two hundred crabs were measured and four groups suspected in advance. Four projections of the data, three found by using projection pursuit with different projection indices, are shown in Figure 2.13. Only the projection in Figure 2.13(b) matches up with the *a priori* expectation of four groups.

Figure 2.13 Results from projection pursuit applied to 200 specimens of the rock crab genus*Leptograpsus*: (a) random projection; (b) from projection pursuit with the Hermite index; (c) from projection pursuit with the Friedman–Tukey index; (d) from projection pursuit with the Friedman (1987) index.

(Reproduced with permission of Cambridge University Press, from Pattern Recognition and Neural Networks, 1996, B. D. Ripley.)

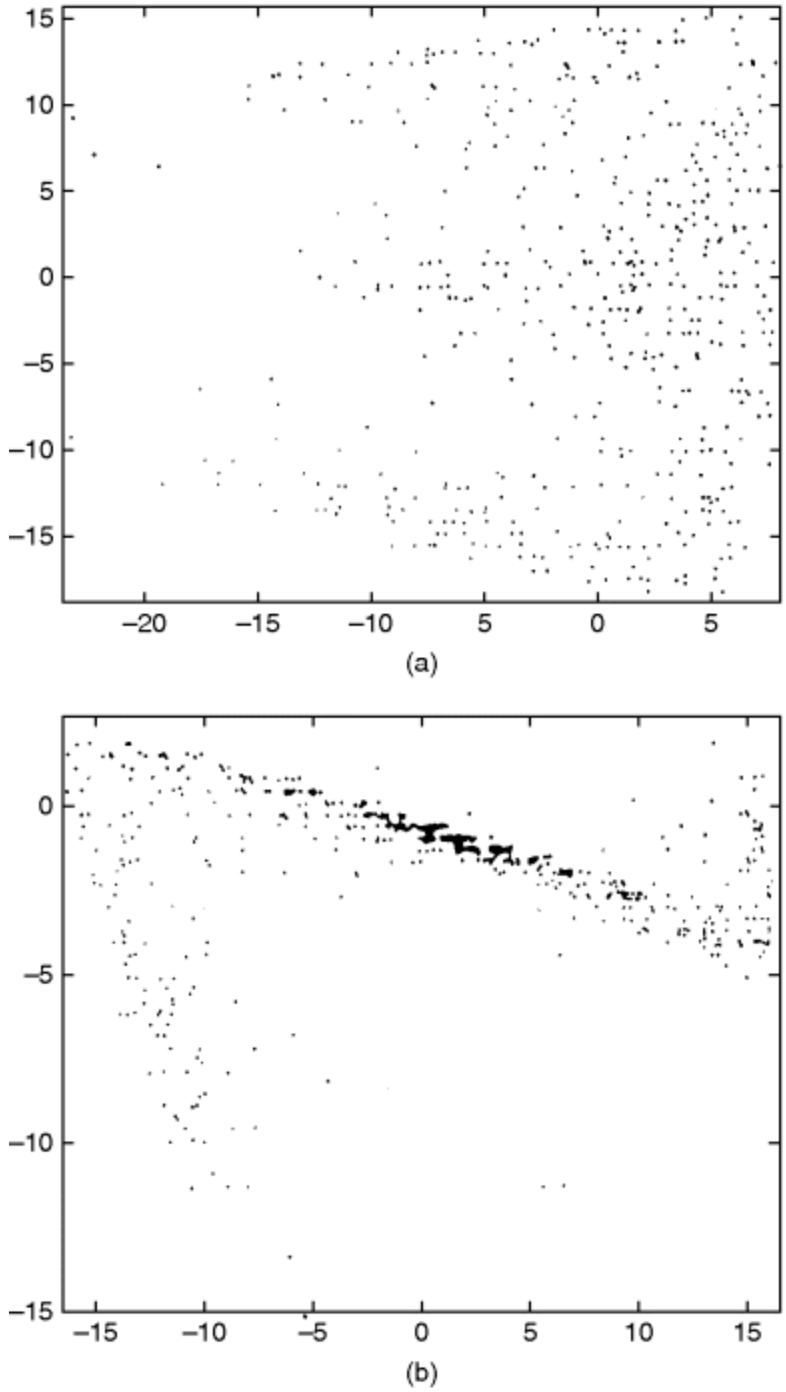


(a) (b) (c) (d)

A further example is provided by Friedman and Tukey (1974). The data consists of 500 seven-dimensional observations taken in a particle physics experiment. Full details are available in the original paper. The scatterplots in Figure 2.14 show projections of the data onto the first two principal components and onto a plane found by projection pursuit. The projection pursuit solution shows structure not apparent in the principal components plot.

Figure 2.14 Two-dimensional projections of high-energy physics data: (a) first two principal components; (b) projection pursuit.

(Reproduced from Friedman and Tukey (1974) with the permission of the Institute of Electrical and Electronic Engineers, Inc.)



Other interesting examples are given in Posse (1995). An account of software for projection pursuit is given in Swayne *et al*. (2003), and a more recent account of the method is given in Su *et al*. (2008).

2.3.3 Multidimensional Scaling

It was mentioned in Chapter 1 that the first step in many clustering techniques is to transform the $n \times p$ multivariate data matrix **X** into an $n \times n$ matrix, the entries of which give measures of distance, dissimilarity or similarity between pairs of individuals. (Chapter 3 will be concerned with details of these concepts.) It was also mentioned that such distance or dissimilarity matrices can arise directly, particularly from psychological experiments in which human observers are asked to judge the similarity of various stimuli of interest. For the latter in particular, a well established method of analysis is some form of *multidimensional scaling* (MDS). The essence of such methods is an attempt to represent the observed similarities or dissimilarities in the form of a geometrical model by embedding the stimuli of interest in some coordinate space so that a specified measure of distance, for example *Euclidean* (see Chapter 3), between the points in the space represents the observed proximities. Indeed, a narrow definition of multidimensional scaling is the search for a low-dimensional space in which points in the space represent the stimuli, one point representing each stimulus, such that the distances between the points in the space d_{ij} match as well as possible, in some sense, the original dissimilarities δ_{ij} or similarities s_{ij}. In a very general sense this simply means that the larger the observed dissimilarity value (or the smaller the similarity value) between two stimuli, the further apart should be the points representing them in the geometrical model. More formally, distances in the derived space are specified to be related to the corresponding dissimilarities in some simple way, for example linearly, so that the proposed model can be written as follows:

$$\delta_{ij} = f\left(d_{ij}\right)$$

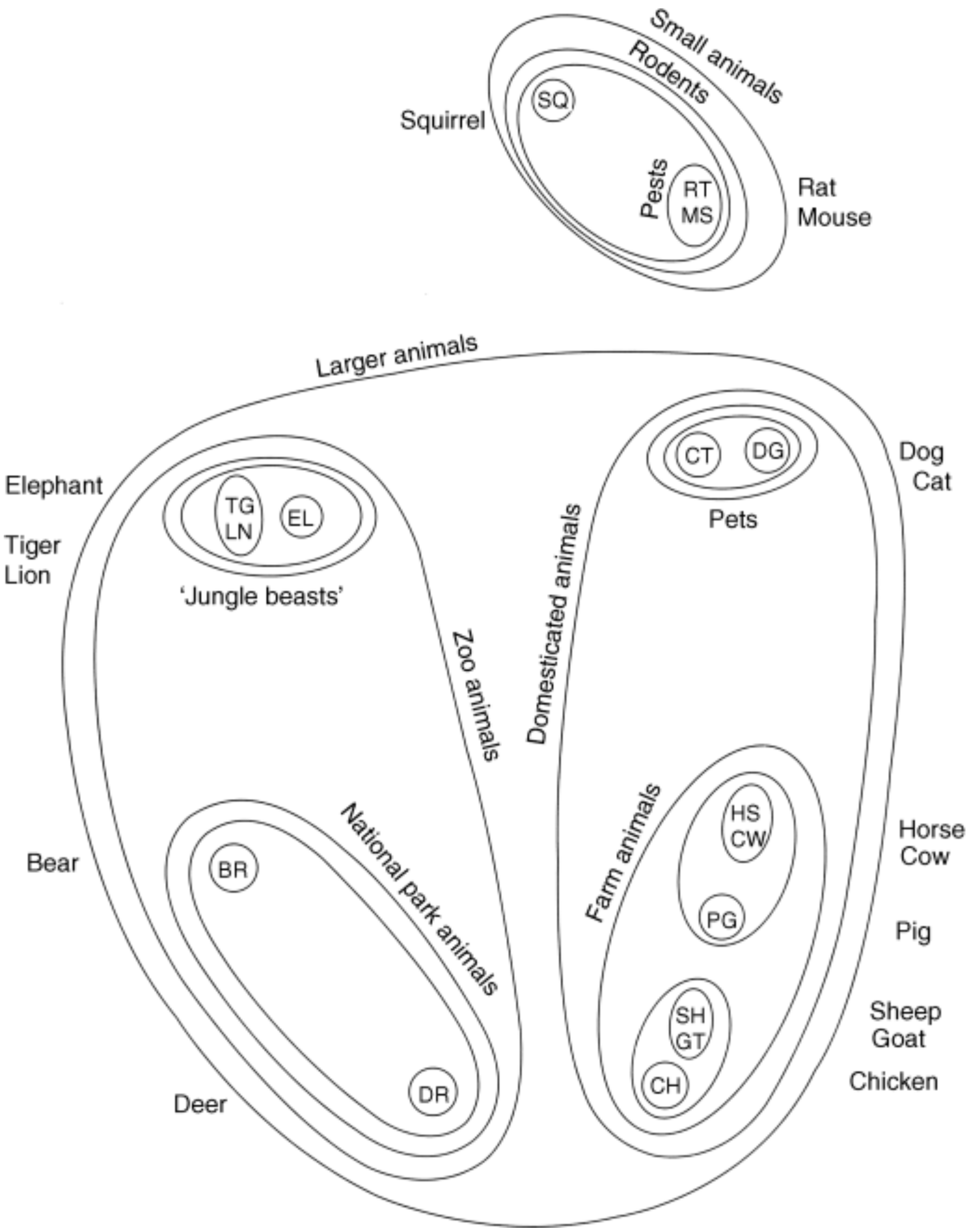
$$d_{ij} = h\left(\mathbf{x}_i, \mathbf{x}_j\right), \quad \textbf{(2.14)}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are q-dimensional vectors ($q < p$) containing the coordinate values representing stimuli i and j, f represents the assumed functional relationship between the observed dissimilarities and the derived distances, and h represents the chosen distance function. In the majority of applications of MDS, h is taken, often implicitly, to be Euclidean. The problem now becomes one of estimating the coordinate values to represent the stimuli. In general this is achieved by optimizing some goodness-of-fit index measuring how well the fitted distances match the observed proximities – for details see Everitt and Rabe-Hesketh (1997).

The general aim of MDS is most often to uncover the dimensions on which human subjects make similarity judgements. But it can also be helpful in clustering applications for displaying clusters in a two-dimensional space so that they can be visualized. An example from Shepard (1974) will illustrate how. In investigating the strengths of mental associations among 16 familiar kinds of animals, Shepard started with the quantitative information to form a MDS solution and subsequently obtained typical information by performing a hierarchical cluster analysis (see Chapter 4). Shepard gained substantially increased insight into the data structure after superimposing the typical information on the MDS 'map' of the data by enclosing cluster members within closed boundaries – see Figure 2.15.

Figure 2.15 Mental associations among 16 kinds of animals. MDS solution plus hierarchical clustering solution

(reproduced with permission from Shepard,1974.)



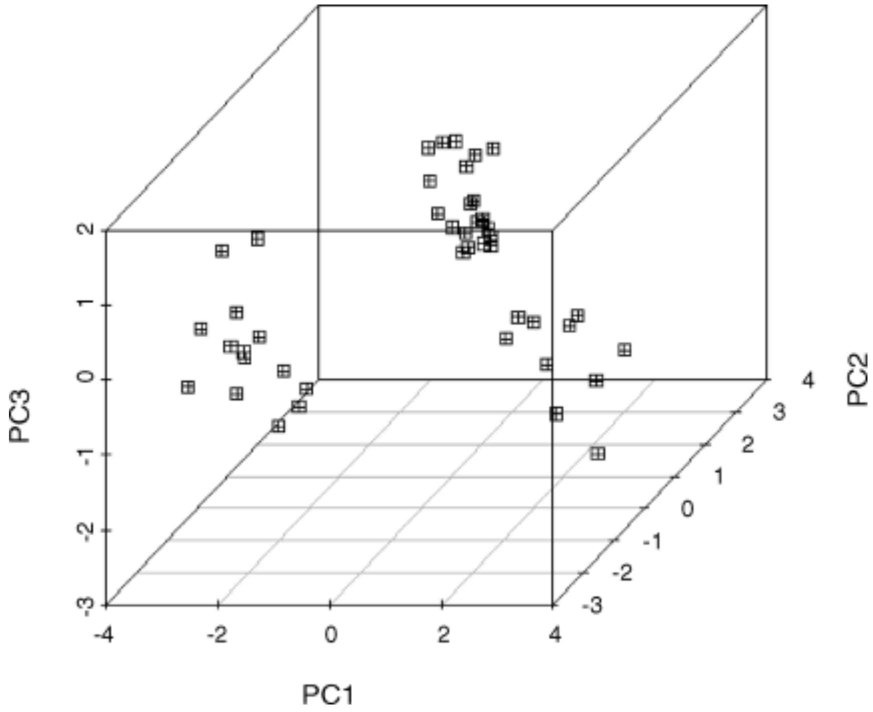
Another example of the use of MDS for the clustering of fragrances is given in Lawless (1989), and a further recent example is described in Adachi (2002).

2.4 Three-dimensional Plots and Trellis Graphics

The previous sections have primarily dealt with plotting data in some two-dimensional space using either the original variables, or derived variables constructed in some way so that a low-dimensional projection of the data is informative. But it is possible to plot more than two dimensions directly, as we will demonstrate in this section by way of a number of examples.

First, three-dimensional plots can now be obtained directly from most statistical software and these can, on occasions, be helpful in looking for evidence of cluster structure. As an example, Figure 2.16 shows a plot of the Romano-British pottery data in the space of the first three principal components. The plot very clearly demonstrates the presence of three separate clusters, confirming what was found previously in Section 2.3.1 using scatterplot matrices and estimated bivariate densities.

Figure 2.16 Three-dimensional plot of the Romano-British pottery data in the space of their first three principal component scores.

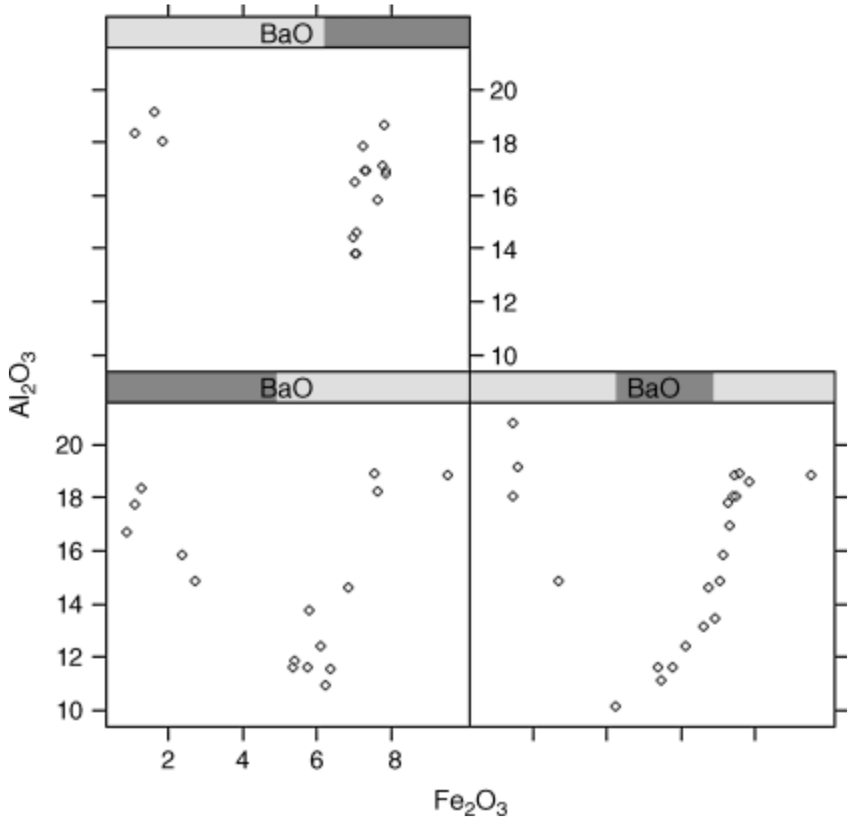


Trellis graphics (Becker and Cleveland, 1994) and the associated lattice graphics (Sarkar, 2008) give a way of examining high-dimensional structure in data by means of one-, two- and three-dimensional graphs. The quintessential feature of the approach is *multiple conditioning*, which allows some type of plot to be displayed for different values of a given variable or variables. Although such graphics are not specifically designed to give evidence of clustering in multivariate data, they might, nevertheless, be helpful for this is some situations. As a simple example of a trellis graphic we will look at scatterplots of Al_2O_3 values against Fe_2O_3 values conditioned on the values of BaO, after dividing these values in to three groups as follows:

Intervals	Min	Max	Count
1	0.0085	0.0155	16
2	0.0145	0.0185	22
3	0.0175	0.0236	16

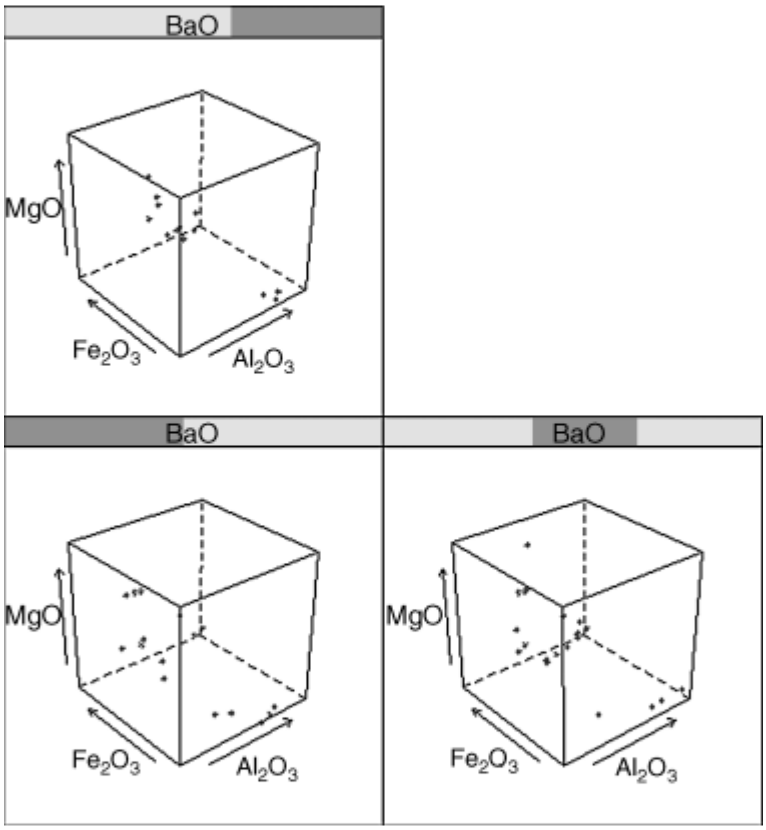
The resulting trellis diagram is shown in Figure 2.17. The evidence of cluster structure in the scatter plots of Al_2O_3 against Fe_2O_3 appears to hold for all values of BaO.

Figure 2.17 Trellis graphic of three variables from the Romano-British pottery data.



A more complex example of a trellis graphic is shown in Figure 2.18. Here three-dimensional plots of MgO against Al_2O_3 and Fe_2O_3 are conditioned on the three levels of BaO used previously. For the lowest values of BaO, the separation of the points into clusters appears to be less distinct than for the medium and higher values.

Figure 2.18 Trellis graphic of four variables from the Romano-British pottery data.



Many other exciting examples of the use of trellis graphics, although not specifically for evidence of clustering, are given in Sarkar (2008).

2.5 Summary

Evidence for the presence of clusters of observations can often be found by some relatively simple plots of data, enhanced perhaps by appropriate density estimates. In one or two dimensions, for example, clusters can sometimes be identified by looking for separate modes in the estimated density function of the data. Such an approach can be used on data sets where the number of variables is greater than two, by first projecting the data into a lower dimensional space using perhaps principal components or a projection pursuit approach. Although in this chapter nonparametric density estimation has been used in an informal manner to identify modes in univariate and bivariate data, Silverman (1986) describes how it can be used as the basis of a more formal approach to clustering. When only a proximity matrix is available, multidimensional scaling can be used in an effort to detect cluster structure visually. And three-dimensional plots and trellis graphics may also, on occasions, offer some graphical help in deciding whether a formal cluster analysis procedure might usefully be applied to the data. (In this chapter, our concern has been graphical displays that may help in deciding whether or not the application of some formal method of cluster analysis to a data set might be worthwhile; in Chapter 9 we will look at some graphics that may be helpful in displaying the clusters obtained from particular clustering techniques.)